QuickCompany.in

Login

< Back

"A Process For Preparation Of ...

Council Of Scientific & Industrial Research

Info | Spec | Docs 21 | ERegister

"A Process For Preparation Of Radical Scavenging Water Soluble Yellow Colourant Turmeric Formulation"

Abstract: Turmeric (roots of Curcuma longa - Zingiberaceae), has been used in traditional medicinal formulations for various treatments which include jaundice and other liver ailments, ulcers, parasitic infections, skin diseases, sprains, inflammation of the joints, cold and flu symptoms. In the present invention turmeric oleoresin, a gummy substance subjected to solvent extraction for removal of the volatile oil as well as the fixed oil to enrich radical scavenging compounds viz., the curcuminoids. The resultant product, called as Curcumin Enriched Turmeric Oleoresin (CETO) powder, is free flowing in nature and brownish - yellow in colour. It is dissolved in a suitable solvent and converted into water-soluble yellow colourant formulation using food additives. The resultant formulation possesses 65-70% radical scavenging activity at 100 ppm and also contains curcuminoids in the range of 10-16%. Hence

Trademark Agent

QuickCompany.in

Login

Get Free WhatsApp Updates!
Notices, Deadlines & Correspondence

FOLLOW

Patent Information

Application # 409/DEL/2009	Filing Date 03 March 2009
Publication Number 37/2010	Publication Type INA
Invention Field BIO-MEDICAL ENGINEERING	Status Granted
Email	Parent Application
Patent Number 362338	Legal Status Inforce
Grant Date 2021-03-22	Renewal Date 03 Mar 2027

Applicants

1. COUNCIL OF SCIENTIFIC & INDUSTRIAL RESEARCH
ANUSANDHAN BHAWAN, RAFI MARG, NEW DELHI - 110 001, INDIA.

Inventors

1. L. JAGAN MOHAN RAO

QuickCompany.in

Login

2. S. NAGARAJAN
PPSFT DEPT. CENTRAL FOOD TECHNOLOGICAL RESEARCH INSTITUTE, MYSORE - 570 020, INDIA.

Specification

The present invention relates to a process for preparation of radical scavenging water-soluble yellow colourant turmeric formulation.
Turmeric (roots of Curcuma longa - Zingiberaceae), has been used in traditional medicine for various treatments which include jaundice and other liver ailments, ulcers, parasitic infections, skin diseases, sprains, inflammation of the joints, cold and flu symptoms. It is extensively used as a colourant and preservative in food products. The major non-volatile compounds of turmeric are curcuminoids (viz., curcumin, demethoxycurcumin and bisdemethoxycurcumin). These exhibit a variety of beneficial effects on health and help in preventing certain diseases. The most detailed studies using curcumin (diferuloyl methane - the major constituent) include anti-inflammatory, antioxidant, anticarcinogenic, antiviral, and antiinfectious activities. In addition, the wound healing and detoxifying properties of curcumin have also received considerable attention. Studies on the therapeutic properties of curcumin revealed some understanding on the cellular, molecular, and biochemical mechanism of action.
Solvent extraction method was found to be most effective for isolation of oleoresin from turmeric. It contains three diaryl heptanoids as major compounds as mentioned above. Curcuminoids, the natural pigments, isolated from turmeric and is widely used as a colouring agent in food, drugs and Cosmetics. It is being used under oil / fat based food products and formulations and not under aqueous medium due to its poor solubility in water. Curcumin is usually the main constituent. In rare cases the other two compounds amounts to nearly 70% of the total curcuminoids. Commecrcially available 'pure' Curcumin consists of the mixture of the three naturally occurring curcuminoids with curcumin as main constituent. It is known that

QuickCompany.in Login

water, but soluble in alkali or glacial acetic acid. In solution at pH 1-7 the colour is yellow and in pH range 7-9 the colour is brownish-red or deep red.

Thus, the isolation of the radical scavenging conserve from turmeric oleoresin (which possess the potential bioactive constituents) was undertaken and converted it to water-soluble yellow colourant formulation, which can readily disperse into water so that it can be effectively used in oil-free foods and beverages as nutraceutical.

The significance of the new process and the products lies in that the turmeric oleoresin provides the potential radical scavenging conserve, which is enriched with curcuminoids content and also has the water-soluble properties.

Reference may be to the US patent No.US 4263333 (1981) entitled 'Curcumin -metal colour complexes' wherein the preparation of a water soluble complex formed from curcurmn using metal complexes for use in foods was described. The drawbacks are the toxicity of metals at higher intakes. Further, the references may be made to two more patents such as UK-GB2 132205A (1984) entitled 'Colouring agents' wherein the preparation of a water-soluble curcumin complex using source of curcumin and gelatin in acetic acid for food applications was revealed and an US patent No. US4368208 (1983) entitled 'Water soluble curcumin 84%' wherein the method for preparing a water- soluble curcumin - gelatin complex colouring agent for foods was described. The drawback is the use of acid in the formulation. In addition to this, a reference was also made to the US patent No. US 4999205 (1991) entitled ' a curcumin complex with a water-soluble branched chain or cyclic polysaccharide or a water-soluble or a water dispersible protein was described. The drawbacks are it involves the use of acids/bases for altering the pH.

Reference may also be made to the CFTRI patent entitled 'The Preparation of the Water soluble Curcumin Formulation (Liquid And Solid) and its Use as Colourant in

Foods and Beverages (401/DEL/01)' wherein the CRTO and the isolated pure curcuminoids were used for the preparation of formulations.

The main object of the present invention is to provide the radical scavenging water soluble yellow colourant turmeric formulation

QuickCompany.in Login

preparation of a turmeric formulation having radical scavenging activity, which comprises the following steps of: (i) dissolving the turmeric oleoresin with a low polar solvent, such as herein described, at a ratio 1: 10-30,
(ii) separating the dissolved fraction by conventional filtration to obtain curcumin enriched turmeric oleoresin (CETO) conserve as solid / powdered material, (iii) mixing the separated solid / powder of step (ii) in a polar solvent as herein described at a ratio 1: 2-5 to get a paste ,
(iv) adding to the above paste an emulsifier selected from the group consisting of polyoxyethylene sorbitan monopalmitate, polyoxyethylene sorbitan monolaurate, polyoxyethylene sorbitan monosterate, polyoxyethylene sorbitan trioleate, gums and glcerides of fattyacids at a ratio 3: 1-4 , stirring for 2-3 hours to get the desired formulation having radical scavenging activity.
In an embodiment of the present invention, the low polar solvent used in step (i) may be low molecular weight hydrocarbon or mixture of hydrocarbons [C5-C8]. In another embodiment of the present invention in step (i), the ratio of material to solvent may be 1:10-30 (or ten-thirty times).
In yet another embodiment of the present invention in step (ii), the filtration set used may be Buchner flask fitted with filter paper having the provision for vacuum application.

In still another embodiment of the present invention, the yield of radical scavenging
fraction (or) solid in step (ii) may be in the range of 40-50%.
In still another embodiment of the present invention, the radical scavenging activity of
the solid obtained in step (ii) may be in the range of 80-85% at 25-ppm level.
In still yet another embodiment of the present invention, the curcuminoid content of the
radical scavenging fraction in step (ii) may be may be in the range of 55-60%.
In still yet another embodiment of the present invention, the polar solvent used to
dissolve the radical scavenging fraction in step (v) may be a dihydric alcohol preferably
selected from the group consisting of [C3-C5].
In still yet another embodiment of the present invention in step

QuickCompany.in Login

In still yet another embodiment of the present invention in step (iv) the emulsifier used
used is preferably selected from the group consisting of polyoxyethylene sorbitan
monopalmitate, polyoxyethylene sorbitan monolaurate, polyoxyethylene sorbitan
monosterate, polyoxyethylene sorbitan trioleate, gums and glcerides of fattyacids.
In still yet another embodiment of the present invention in step (iv) the quantity of
emulsifier used in the ratio of solution to emulsifier may be 3: 1-4 (one-four times).
In still yet another embodiment of the present invention the radical scavenging activity
of the formulation obtained in step (iv) may be in the range of 65-70% at 100 ppm level.
In still yet another embodiment of the present invention in step (vi), the curcuminoid
content of the radical scavenging and water-soluble formulation obtained in step (vi) may
be in the range of 10-16%.
(Formula Removed)
The novelty of the new process lies in the preparation of potential and natural radical scavenging conserve powder from the oleoresin of turmeric and converting it to water soluble, natural yellow colourant, odourless formulation, which can serve dual purpose (viz., nutraceutical as well as colouring agent) in food systems.
Novel inventive steps involve, separation of radical scavenging conserve powder by removal of undesirable material through solvent extraction from turmeric oleoresin and its conversion to water-soluble formulation.
The following examples are given by way of illustration of the present invention and therefore should not be construed to limit the scope of the present invention.
Example-1
Turmeric oleoresin, a gummy substance 10 g taken in a beaker and 250 ml of hexane was added and stirred for 4 h. The volatile oil as well as the fixed oil from the oleoresin got dissolved in the solvent and the precipitate having yellow brownish colour was separated. The supernatant was decanted and the residue

QuickCompany.in Login

desiccator over P2O5 and Paraffin wax to remove the residual solvent and moisture present in it. The resultant product, called as Curcumin Enriched Turmeric Oleoresin (CETO) powder, is free flowing in nature and brownish - yellow in colour (yield, 5g). The curcuminoids content of the CETO powder was found to be 56±2%. Radical scavenging activity of the CETO powder was found be in the range of 81 ±1% at 25-ppm concentration using DPPH method. CETO powder 1 g was taken and 2 g of propylene glycol was added and made into paste. 2 g of Polyoxyethylene (20) sorbitan mono-oleate (PSM, a non-ionic surfactant) was added and stirred for 2 h. to get a viscous liquid of deep red colour i.e., radical scavenging water soluble yellow colourant formulation. The curcuminoids content in this formulation was found to be 11.5%. Radical scavenging activity of this formulation was found be in the range of 68±1% at 100 ppm concentration Example-2

Turmeric oleoresin, a gummy substance 20 g taken in a beaker and 400 ml of hexane was added and stirred for 5 h. The volatile oil as well as the fixed oil from the oleoresin got dissolved in the solvent and the precipitate having yellow brownish colour was separated. The supernatant was decanted and the residue filtered and thoroughly washed with solvent. The residue was air dried to get free flowing powder and then kept in a vacuum desiccator over P1O5 and Paraffin wax to remove the residual solvent and moisture present in it. The resultant product, called as Curcumin Enriched Turmeric Oleoresin (CETO) powder, is free flowing in nature and brownish - yellow in colour (yield, 9.5g). The curcuminoids content of the CETO powder was found to be 55±3%. Radical scavenging activity of the CETO powder was found be in the range of 80+2% at 25 ppm concentration. CETO powder 2g was taken and 4.5g of propylene glycol was added and made into paste. 4.5 g of Polyoxyethylene (20) sorbitan mono-oleate (PSM, a non-ionic surfactant) was added and stirred for 2 h. to get a viscous liquid of deep red colour i.e., radical scavenging water soluble yellow colourant formulation. The curcuminoids content in this formulation was found to be 10.5%. Radical scavenging activity of this formulation was found be in the range of 67 ±2% at 100 ppm concentration Example -3

Lemon juice (75 ml) obtained from fresh fruits of lemon was taken in a beaker and added 2 L of water. To this 200 g of Sugar was added and stirred. 25 mg of formulation was added to 2 L of

QuickCompany.in Login

satisfactory.
To determine the curcumin content a standard method of measuring absorbance with UV spectrophotmeter at 425 nm was used. The solution 0.1 ml was taken and diluted to 10,000 times with ethanol and the absorbance (UV) was measured. Based on this , the curcuminoids content was calculated the formulation and was found out to be 10-13%. These formulations, when added to water (25mg in 2L), give a clear solution of pleasing yellow colour. Example - 4
The liquid formulation prepared as above is converted into powder as follows. 10-20 parts of the above formulation is mixed with 10-100 parts of Carbohydrate monomer/polymer, and the impregnated solid dried in dessicator to get solid formulation. The colour value of the formulations is determined by the UV-Spectroscopic method as described earlier. The curcuminoids content in the formulations was found to be 1-4 %. Example - 5
The fruit juice (75 ml) obtained from fruits (such as lime, lemon, pineapple etc) was taken in a beaker and mixed with 2 L of water. To this 200 g of Sugar was added and stirred. The formulation one drop (25mg) dissolved in 20 ml of water and added to 2 L of beverage and stirred well. The product was subjected to sensory evaluation and found to be satisfactory. The coloured beverage was taken in 200 ml bottles and kept at room temperature and in Refrigerator (4-6°C). The main advantages of the present invention are:
1. The formulations described are useful as natural yellow colourant as well as
nutraceutical in several food products such as ice creams, bakery-products, snacks
(including extruded products), desserts, beverages and confectionery.
2. The amount of formulation required to obtain pleasing yellow colour as well as radical
scavenging activity for various food products is at ppm levels.
3. The formulations prepared are odour free.

We claim
1. A process for radical scavenging water-soluble yellow colourant turmeric formulation
which comprises the steps of:
(i) dissolving the turmeric oleoresin with a low polar solvent, such as herein described, at

QuickCompany.in Login

obtain curcumin
enriched turmeric oleoresin (CETO) conserve as solid / powdered material,
(iii) mixing the separated solid / powder of step (ii) in a polar solvent as herein described
at a ratio 1: 2-5 to get a paste ,
(iv) adding to the above paste an emulsifier selected from the group consisting of
polyoxyethylene sorbitan monopalmitate, polyoxyethylene sorbitan monolaurate,
polyoxyethylene sorbitan monosterate, polyoxyethylene sorbitan trioleate, gums and
glcerides of fattyacids at a ratio 3: 1-4 , stirring for 2-3 hours to get the desired
formulation having radical scavenging activity.

2. A process as claimed in claim 1 wherein, the low polar solvent used at step (i) is
selected from low molecular weight hydrocarbon or mixture of hydrocarbons [C5-C8]
such as hexane.

3. A process as claimed in claims 1-2 wherein, the conventional filtration used at step (ii)
is being filtration through Buchner flask fitted with filter paper and having the provision
for vacuum application.

4. A process as claimed in claims 1-3 wherein the yield of radical scavenging fraction
obtained at step (ii) is in the range of 40-50%.

5. A process as claimed in claims 1-4 wherein the curcuminoid content of the radical
scavenging fraction obtained at step (ii) is in the range of 55-60%.

6. A process as claimed in claims 1-5 wherein the polar solvent used to dissolve the
radical scavenging fraction at step (iii) is selected from dihydric alcohol preferably
selected from the group consisting of [C3-C5] such as propylene glycol.

7. A process as claimed in claims 1-6 wherein the radical scavenging activity of the
formulation obtained in step (iv) is in the range of 65-70% at 100

QuickCompany.in Login

yellow colourant
turmeric formulation substantially as herein described with reference to the examples.

Documents

APPLICATION DOCUMENTS

#	Name	Date
1	409-del-2009-Form-18-(20-06-2011).pdf	2011-06-20
2	409-del-2009-Correspondence Others-(20-06-2011).pdf	2011-06-20
3	409-del-2009-form-5.pdf	2011-08-21
4	409-del-2009-form-3.pdf	2011-08-21
5	409-del-2009-form-2.pdf	2011-08-21
6	409-del-2009-form-1.pdf	2011-08-21
7	409-del-2009-description (complete).pdf	2011-08-21
8	409-del-2009-correspondence-others.pdf	2011-08-21
9	409-del-2009-claims.pdf	2011-08-21
10	409-del-2009-abstract.pdf	2011-08-21
11	409-DEL-2009-FER.pdf	2017-07-20
12	409-DEL-2009-OTHERS [18-01-2018(online)].pdf	2018-01-18
13	409-DEL-2009-FER_SER_REPLY [18-01-2018(online)].pdf	2018-01-18
14	409-DEL-2009-COMPLETE SPECIFICATION [18-01-	2018-01-18

15	409-DEL-2009-CLAIMS [18-01-2018(online)].pdf	2018-01-18
16	409-DEL-2009-Response to office action (Mandatory) [20-06-2018(online)].pdf	2018-06-20
17	409-DEL-2009-Annexure (Optional) [20-06-2018(online)].pdf	2018-06-20
18	409-DEL-2009-NBA Approval Submission [22-09-2020(online)].pdf	2020-09-22
19	409-DEL-2009-PatentCertificate22-03-2021.pdf	2021-03-22
20	409-DEL-2009-IntimationOfGrant22-03-2021.pdf	2021-03-22

SEARCH STRATEGY

1	409searchstrgy_18-07-2017.pdf

ERegister / Renewals Inforce

3RD: 21 JUN 2021

CBR 21665 | Renewal 21/06/2021 | Renewal Amount ₹4,000 | Certificate #43143

From 03/03/2011 - To 03/03/2012

4TH: 21 JUN 2021

CBR 21665 | Renewal 21/06/2021 | Renewal Amount ₹4,000 | Certificate #43144

From 03/03/2012 - To 03/03/2013

QuickCompany.in Login

CBR 21665 | Renewal 21/06/2021 | Renewal Amount ₹4,000

Certificate #43145

From 03/03/2013 - To 03/03/2014

6TH: 21 JUN 2021

CBR 21665 | Renewal 21/06/2021 | Renewal Amount ₹4,000

Certificate #43146

From 03/03/2014 - To 03/03/2015

7TH: 21 JUN 2021

CBR 21665 | Renewal 21/06/2021 | Renewal Amount ₹12,000

Certificate #43147

From 03/03/2015 - To 03/03/2016

8TH: 21 JUN 2021

CBR 21665 | Renewal 21/06/2021 | Renewal Amount ₹12,000

Certificate #43148

From 03/03/2016 - To 03/03/2017

9TH: 21 JUN 2021

CBR 21665 | Renewal 21/06/2021 | Renewal Amount ₹12,000

Certificate #43149

From 03/03/2017 - To 03/03/2018

10TH: 21 JUN 2021

CBR 21665 | Renewal 21/06/2021 | Renewal Amount ₹12,000

Certificate #43150

From 03/03/2018 - To 03/03/2019

11TH: 21 JUN 2021

QuickCompany.in Login

Certificate #43151

From 03/03/2019 - To 03/03/2020

12TH: 21 JUN 2021

CBR 21665 | Renewal 21/06/2021 | Renewal Amount ₹24,000 | Certificate #43152

From 03/03/2020 - To 03/03/2021

13TH: 21 JUN 2021

CBR 21665 | Renewal 21/06/2021 | Renewal Amount ₹24,000 | Certificate #43153

From 03/03/2021 - To 03/03/2022

14TH: 22 FEB 2022

CBR 6714 | Renewal 22/02/2022 | Renewal Amount ₹24,000 | Certificate #13023

From 03/03/2022 - To 03/03/2023

15TH: 15 FEB 2023

CBR 7113 | Renewal 15/02/2023 | Renewal Amount ₹24,000 | Certificate #14336

From 03/03/2023 - To 03/03/2024

16TH: 21 FEB 2024

CBR 12921 | Renewal 21/02/2024 | Renewal Amount ₹40,000 | Certificate #47364

From 03/03/2024 - To 03/03/2025

17TH: 28 FEB 2025

Certificate #17127

From 03/03/2025 - To 03/03/2026

18TH: 06 FEB 2026

CBR 7607 | Renewal 06/02/2026 | Renewal Amount ₹40,000 | Certificate #11456

From 03/03/2026 - To 03/03/2027